

Programming Language Popularity Trends

Data-backed briefing using TIOBE Index, March 2026

What matters now

- Python remains dominant, but posts a -2.59 percentage-point YoY decline.
- C climbs from #4 to #2 with +2.02 points, while C++ and Java both soften.
- R (+0.94) and Perl (+1.05) are standout risers, signaling niche strength.
- Go records the steepest decline among top 20 (-1.49), while Swift re-enters top 20.

Top Line

Python: 21.25% (#1)
C: 11.55% (#2, +2.02 YoY)
Top 4 share: 48.97%
Go: #7→#16 (-1.49 YoY)

2. Top 20 Rankings (Mar 2026): Concentration and Competitive Positioning

Full top-20 ranking with YoY movement (rank and rating delta)

Rank	2025 Rank	Language	Rating	YoY Δ
1	1	Python	21.25%	-2.59%
2	4	C	11.55%	+2.02%
3	2	C++	8.18%	-2.90%
4	3	Java	7.99%	-2.37%
5	5	C#	6.36%	+1.49%
6	6	JavaScript	3.45%	-0.01%
7	9	Visual Basic	2.50%	-0.02%
8	8	SQL	2.00%	-0.57%
9	16	R	1.88%	+0.94%
10	10	Delphi/Object Pascal	1.80%	-0.36%
11	24	Perl	1.75%	+1.05%
12	12	Scratch	1.63%	-0.03%
13	11	Fortran	1.45%	-0.25%
14	14	Rust	1.31%	+0.09%
15	15	MATLAB	1.29%	+0.31%
16	7	Go	1.29%	-1.49%
17	17	Assembly language	1.29%	+0.42%
18	13	PHP	1.23%	-0.25%
19	18	Ada	1.10%	+0.25%
20	26	Swift	1.04%	+0.44%

Concentration

Top 4 (Python, C, C++, Java) =
48.97% of index

Fast risers

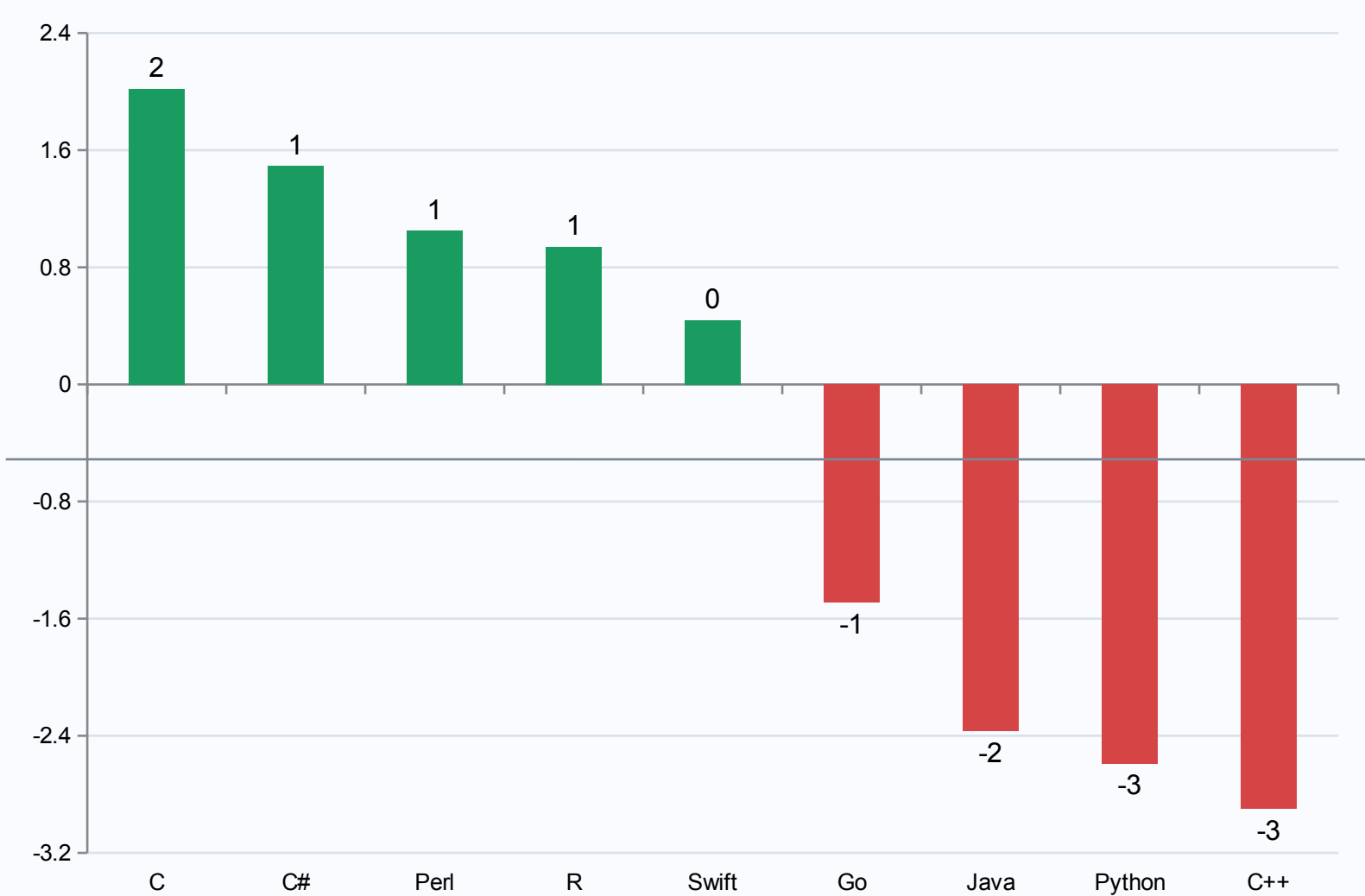
C +2.02
C# +1.49
Perl +1.05
R +0.94

Largest drops

C++ -2.90
Python -2.59
Java -2.37
Go -1.49

3. Biggest Movers YoY: Momentum Signals and Potential Inflection Points

Comparing top positive and negative year-over-year rating shifts



Gainers

C leads gains (+2.02)
C# remains strong (+1.49)
Perl and R post notable rebounds

Losers

Go has steep operational signal (-1.49)
Java and Python decline materially
C++ records largest YoY drop (-2.90)

4. Historical Trends (2002–2026): Structural Shifts in Language Demand

Key structural patterns from the TIOBE top-10 long-run chart

2002–2026 Trend Highlights

- Python rises from around 8th (2006) to #1 after 2018, peaking above 25% before settling at 21.25% in Mar 2026.
- C remains resilient with a top-2 position for most of the 24-year period, including a 2025–2026 surge.
- Java declines from sustained 25%+ leadership (mid-2000s to late-2010s) to under 8% by Mar 2026.
- JavaScript plateaus near ~5% ceiling historically and sits at 3.45% despite web dominance.
- Market concentration persists: leadership changes occur, but top-tier incumbency remains strong.

Implication: Platform Bets

Language strategy should separate cyclical momentum from durable ecosystem depth. Long-term demand curves remain sticky.

Signals to Monitor

- Whether C sustains #2 momentum
- Whether Python stabilizes after decline
- Whether Rust accelerates beyond gradual gains
- Whether Java decline slows or persists

5. Long-Term Resilience (1986–2026): What Persists, What Fades

Cross-decade rank persistence and structural turnover

Language	2026	2021	2016	2011	2006	2001	1996	1991	1986
Python	1	3	5	7	8	26	17	-	-
C	2	1	2	2	2	1	1	1	1
C++	3	4	3	3	3	2	2	2	9
Java	4	2	1	1	1	3	31	-	-
C#	5	5	4	5	7	11	-	-	-
JavaScript	6	7	8	10	10	8	33	-	-
Go	10	11	60	16	-	-	-	-	-
Ada	14	36	25	22	16	18	6	10	3
Lisp	26	34	26	13	14	19	8	4	2

Most Durable

C is the only language remaining in top 2 across all nine measurement points.

Fastest Rise

Python moved from unranked pre-1991 to #1 in 2026 over a 25-year climb.

Long Decline

Lisp fell from #2 (1986) to #26 (2026), illustrating deep ecosystem turnover.

6. Strategic Implications for Engineering Leadership

Actions for portfolio, hiring, architecture, and risk management

Recommended Actions for Engineering Leadership

1) Hedge Core Stacks

Maintain Python productivity advantages while increasing C/C++ proficiency for performance-critical layers.

2) Rebalance Hiring Signals

Track momentum languages (C#, Rust, Swift) alongside incumbents; avoid overreacting to one-year swings.

3) Plan for Legacy Reality

Perl, R, and Fortran resilience implies sustained maintenance demand; preserve institutional expertise.

4) Use Multi-Horizon Governance

Combine quarterly ranking shifts with multi-decade persistence data before major platform decisions.

Bottom line: Treat language rankings as strategic signals, not standalone mandates.